

Stepan
Personal Care

When Sulfate Free is The Economical Solution



What are the main challenges you are facing when formulating Sulfate-free?

Do not edit
How to change the design

The Slido app must be installed on every computer you're presenting from

slido

What matters need to be addressed?

1. Drivers leading to changes in Personal Care Industry
2. Existing Challenges in Meeting Consumer Needs

What insights can we derive from this?

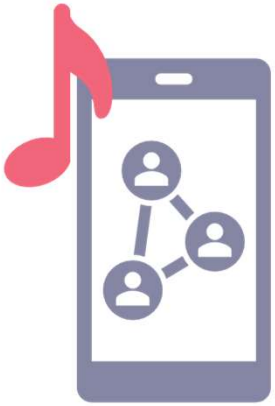
- Popularity Ingredients Growth
- Stepan's Ideal Ingredient Partners for **New Eco-Friendly Habits**
- Innovative Approach to **Transition for a Better Planet**
- Efficient Formulation Techniques for Countless Options

Media, Social Attention Leads to Greater Scrutiny

Consumers may take steps to avoid ingredients receiving negative publicity



Retailers' "Banned Ingredient" Lists



Social Media



Government Regulations

EU Commission Raises Standards through EUDR and 1,4-Dioxane Limit

The palm oil industry and 1,4-Dioxane are facing increased scrutiny

EUDR
European Union Regulation deforestation-free

1,4-Dioxane in EU

Prepare for the Future and Act Now

2021

1,4-Dioxane classified as a Substance of Very High Concern (SVHC)



Danger! According to the **harmonised classification and labelling** (ATP17) approved by the European Union, this substance may cause cancer, is a highly flammable liquid and vapour, causes serious eye irritation and may cause respiratory irritation.

2022

2023

BAUA[1] Call for evidence

Proposal to restrict use of surfactant containing 1,4-Dioxane to **1 ppm on surfactant[2]**

2024

Initial planned implementation Dec 30, 2024

Extra 12 months of phasing-in time for EUDR
30 th December 2025

2025

Oct[3]: 1,4-Dioxane Restriction Submission to ECHA

2026

2027-2028

Entry into force of 1,4-Dioxane Restriction[3]

[1] *BAUA : German Federal Institute for Occupational Safety and Health (Ministerial research institution)*
[2] *There are certain expectations that the actual content of the restriction proposal, including the exact scope and level of the restriction, may actually differ*
[3] *Actual expected regulatory process*

Stepan Personal Care

Turn Challenges into Opportunities

EUDR and 1,4-D Premium Moving Costs Up

Regulation	Palm SLES/SLS[2]	Coconut	Petrochem
EUDR	*Premium ad*	*Premium Exempted[1]*	*Premium Exempted*
1,4-Dioxane	*Premium ad*	*Premium Exempted*	*Premium Exempted*

Recommended Stepan Solutions

NINOL® CAA	BIO-TERGE® AS-40 HASB

[1]*Based on actual EUDR annexe 1*
[2] *SLES: Sodium Lauryl Ether Sulfate; SLS= Sodium Lauryl Sulfate*

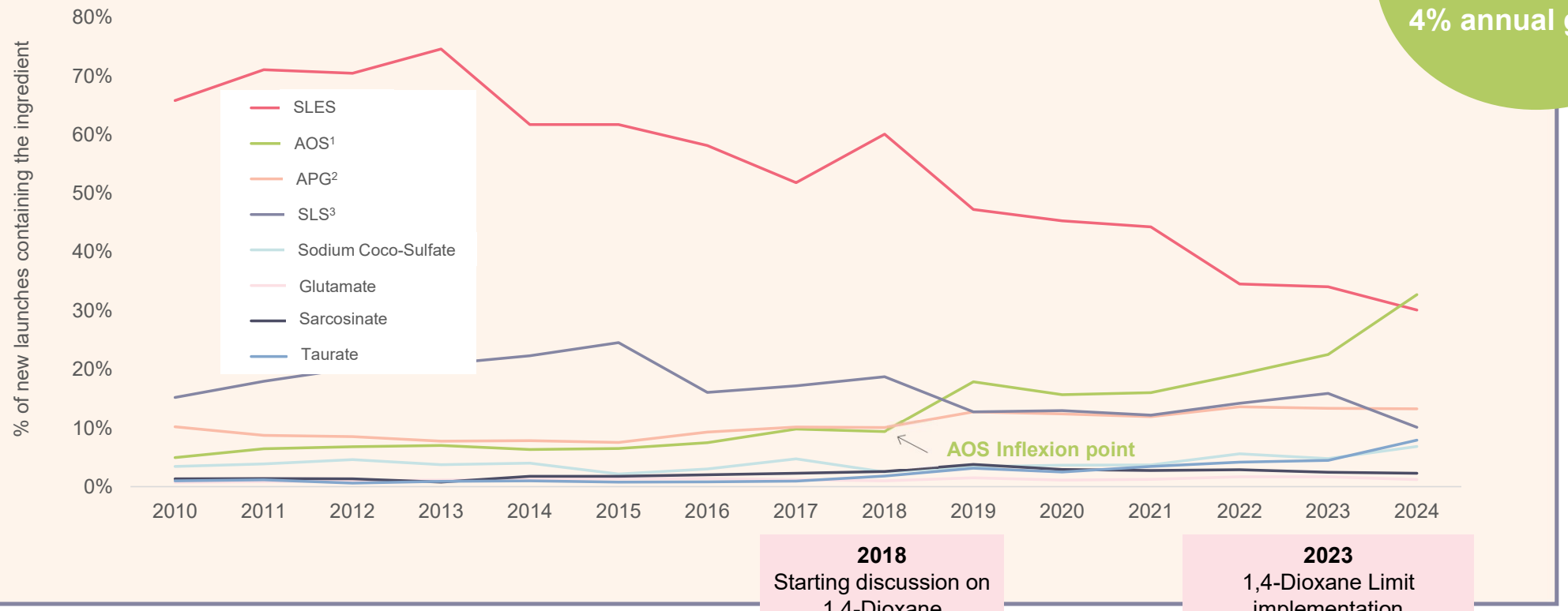
AOS[1] Gaining Momentum: The Formula for Success

Ingredient Popularity Growth Over the Past 15 Years for New Launches

Period 2018-2024

AOS
4% annual growth

Soap & Bath Products; Hair Products [4]: North America



2018
Starting discussion on 1,4-Dioxane

2023
1,4-Dioxane Limit implementation
1 ppm in finished product

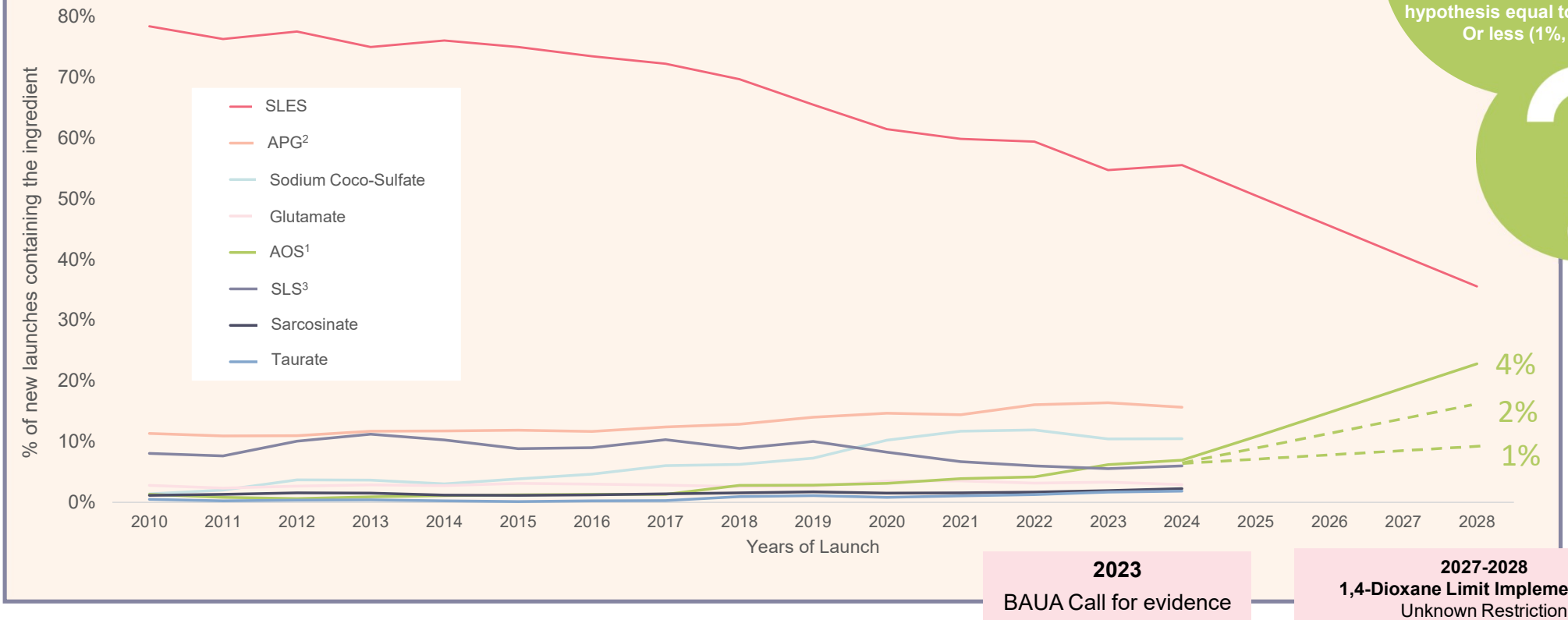
[1]*AOS= Alpha Olefin sulfonate*
[2]*APG= Alkyl Polyglucoside*
[3]*SLS= Sodium Lauryl Sulfate*
[4]*Mintel GNPD*

Stepan Personal Care

AOS[1] Gaining Momentum: The Formula for Success

Ingredient Popularity Growth Over the Past 15 Years for New Launches

Soap & Bath Products; Hair Products [4]: Europe



Projection Period 2024-2028
AOS
4% annual growth
hypothesis equal to NA Growth Or less (1%, 2%...)

2023
BAUA Call for evidence

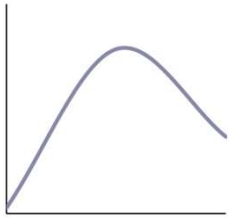
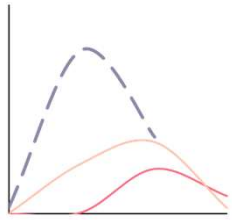
2027-2028
1,4-Dioxane Limit Implementation
Unknown Restriction

[1]AOS= Alpha Olefin sulfonate
[2]APG= Alkyl Polyglucoside
[3]SLS= Sodium Lauryl Sulfate
[4]Mintel GNPD

Stepan Personal Care

Formulating Sulfate Free Cleansers Can Be Challenging

High levels of residual salt, poor viscosity building characteristics, and stability issues can lead to an iterative development process

	Sulfate-Based Systems Primary + Secondary Surfactant	**Modern Sulfate Free Systems** 2+ Co-Surfactants
Viscosity Response		
Clarity, Stability		
Development Time		

Ideal Partners for Tackling Challenges and Guiding Your New Eco-Friendly Habits

BIO-TERGE® AS-40 HASB
INCI: Sodium C14-16 Olefin Sulfonate

AMPHOSOL® DM/MB
INCI: Coco-Betaine

NINOL® CAA
INCI: Dimethyl Lauramide/Myristamide

Stepan Personal Care

Sulfate-Free Alternatives Foamer to SLES

Start Now with Stepan Foamer Solutions

	BIO-TERGE AS-40 HASB INCI: Sodium C14-16 Olefin Sulfonate
Claims	
Natural Origin Index/Content ***ISO 16128 Score***[2]	0.59
Free From	Sulfates, EO, Preservatives, Polymers, Palm
Biodegradability	Readily
Mildness *(Zein, 1-100 Scale, Lower is Better)*	Equal to SLES-2[1]
Leaping Bunny Compliant	✓
Vegan Compliant	✓
Actives %	38,5
Active Use Level %	4-12%
Handling	Clear Amber Liquid at RT
Cost impact	Low
Lab Workup	Back-Add, Adjust pH, and Mix
Category	Mass Personal Care Hair care, Bath & Shower, Bar Soap



[1] *SLES-2: 2-mole Sodium Laureth Sulfate used as a benchmark for comparison purposes. Relative Mildness through Zein test is 58.*
[2] *Natural Origin Index/Content (NOI/NOC) Calculation according to ISO 16128*
Disclaimer: ISO 16128: Stepan interprets the Natural Origin Index (NOI) under ISO 16128 to be calculated based on the biorenewable carbon approach allowed by the norm.
We take water into account (NOI=1), additives and known impurities. Stepan assumes all raw materials and processes used to manufacture our products to be accepted by the norm.

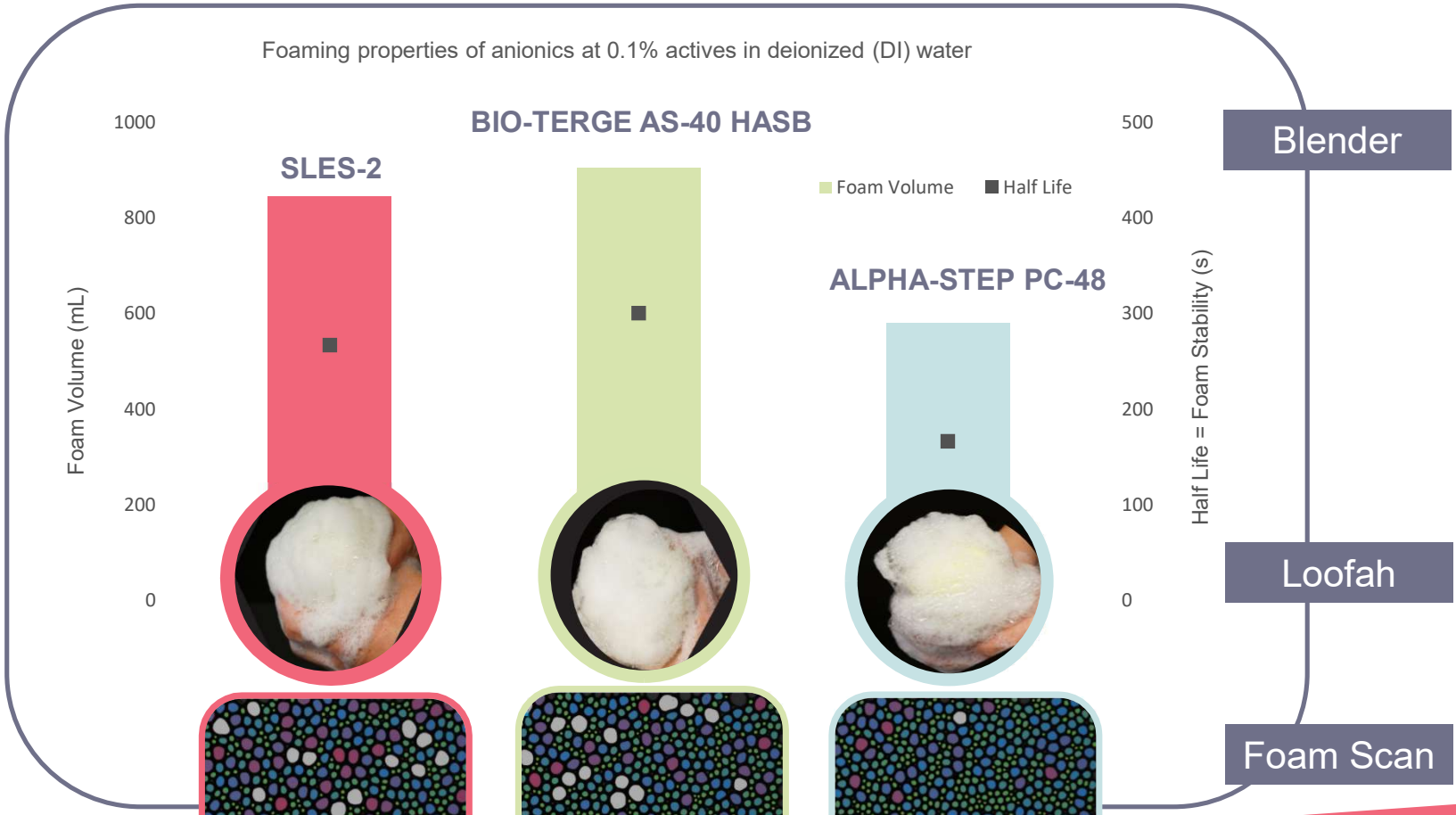
Stepan®
Personal Care

Explore the Bubbly Realm of Sulfonate

Achieve comparable foam levels to SLES or enhance sensory experiences further.

BIO-TERGE AS-40 HASB
Creates foam height similar to SLES-2 with same bubble size.
Produces a high-foaming and loose lather.

ALPHA-STEP PC-48
Lower foam height
Improves sensory experience
Smaller bubble size contributes to a creamier texture.



Stepan Personal Care

Stepan's 6 Formulation "Hacks"

Stepan
Personal Care

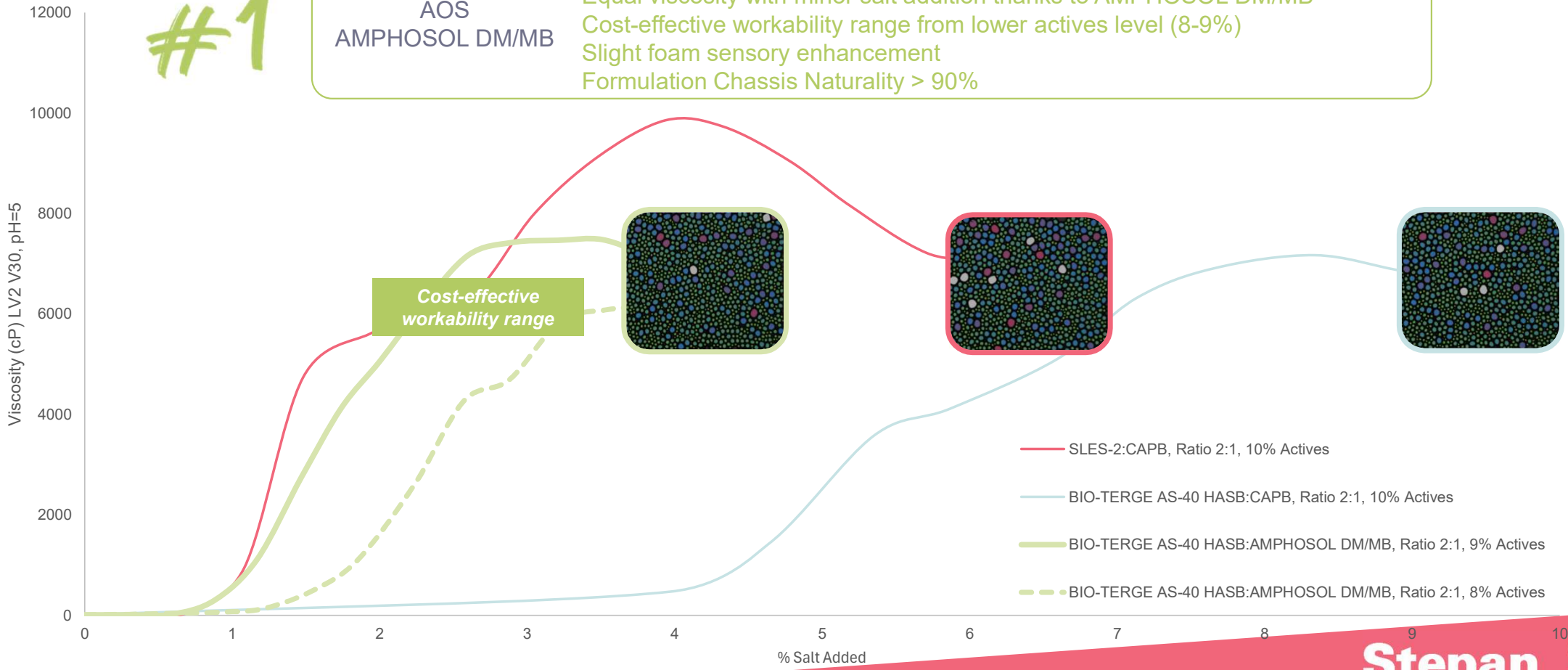
Sulfonate Thickening Hacks

Achieve similar performance to benchmark SLES-2/CAPB while **managing costs**

#1

AOS
AMPHOSOL DM/MB

Equal viscosity with minor salt addition thanks to AMPHOSOL DM/MB
Cost-effective workability range from lower actives level (8-9%)
Slight foam sensory enhancement
Formulation Chassis Naturality > 90%



Stepan Personal Care

Specialty Secondary Surfactants

Amphoteric surfactants, such as COSMOS-certified AMPHOSOL DM/MB and AMPHOSOL CDB-HP-E betaines can provide increased viscosity building and enhanced foam quality while supporting key marketing claims.

AMPHOSOL DM/MB

INCI Name	Coco Betaine
Appearance	Colorless to light yellow liquid
Active	30%
Preservative	None

Verified by ECOCERT
COSMOS APPROVED



Enhances mildness
Antistatic agent

BCI=78
Readily biodegradable
No preservative

Easily thickens sulfate-free systems, even glucosides
Easy handling and formulating
Achieve water-white formulas

AMPHOSOL CDB-HP-E

INCI Name	Cetyl Betaine
Appearance	Colorless to light yellow liquid
Active	30%
Preservative	None



Enhances mildness
Antistatic agent

BCI=78
Readily biodegradable
No preservative

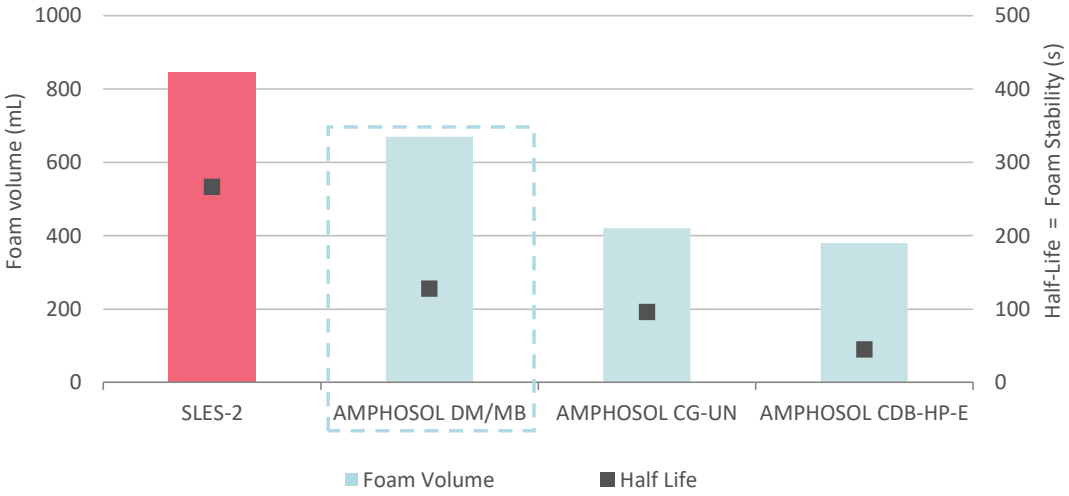
Easily thickens sulfate-free systems, even glucosides
Easy handling and formulating
Achieve water-white formulas

Specialty Secondary Surfactants

Achieve high viscosities and mild sulfate-free formulations by utilizing AMPHOSOL DM/MB and AMPHOSOL CDB-HP-E that can provide excellent thickening with sulfate-free anionic surfactants, better than the standard secondary CAPB.

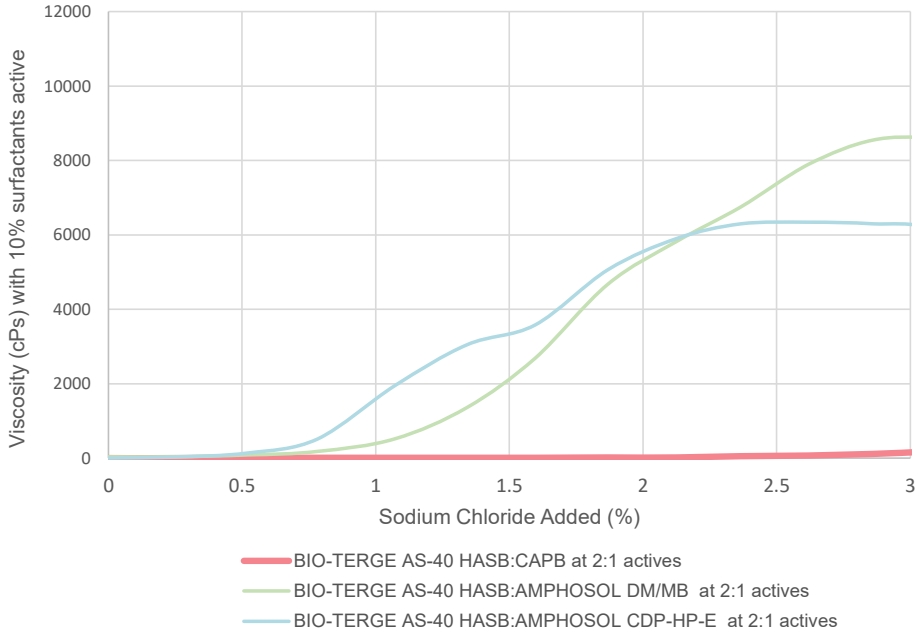
Foam Profiles

AMPHOSOL DM/MB: best candidate for foam



Viscosity Building with Betaines

AMPHOSOL DM/MB and AMPHOSOL CDB-HP-E provides faster viscosity response



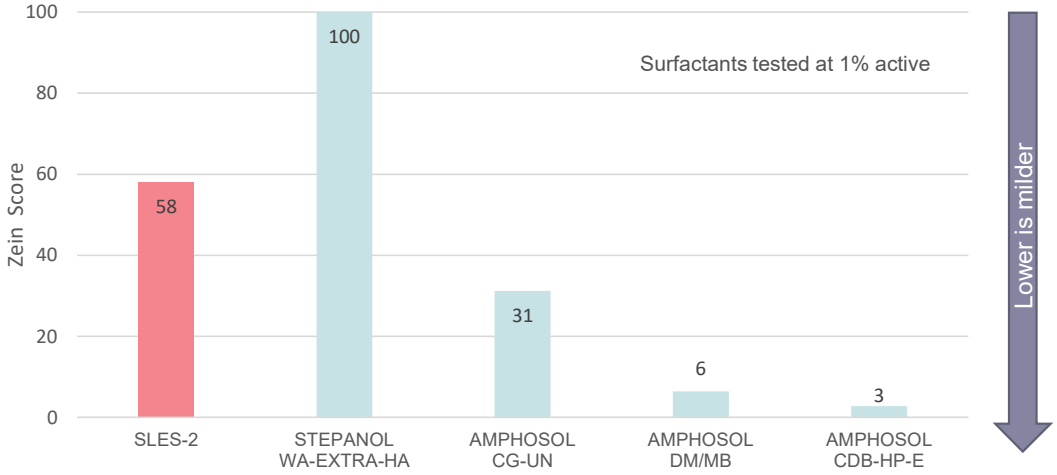
Stepan Specialty Secondary Surfactants

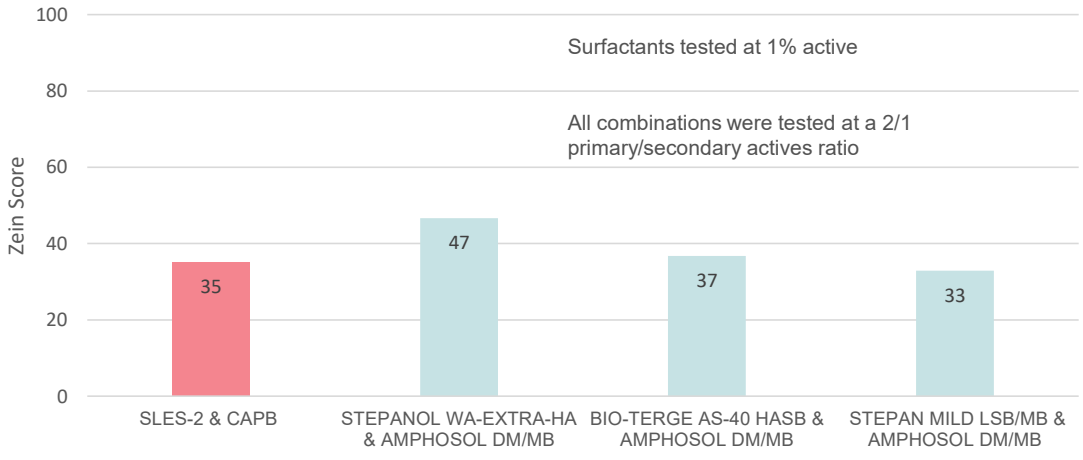
Relative Mildness

The Zein score of a surfactant can be used as an indication of its relative mildness. The lower the Zein score, the milder the surfactant. Scores are normalized to Sodium Lauryl Sulfate, STEPANOL WA-EXTRA-HA, which is given a score of 100.

AMPHOSOL DM/MB, best secondary surfactant for foaming and viscosity performance in economical, mild, sulfate-free formulations

Betaine Zein Score compared to SLES-2 and SLS





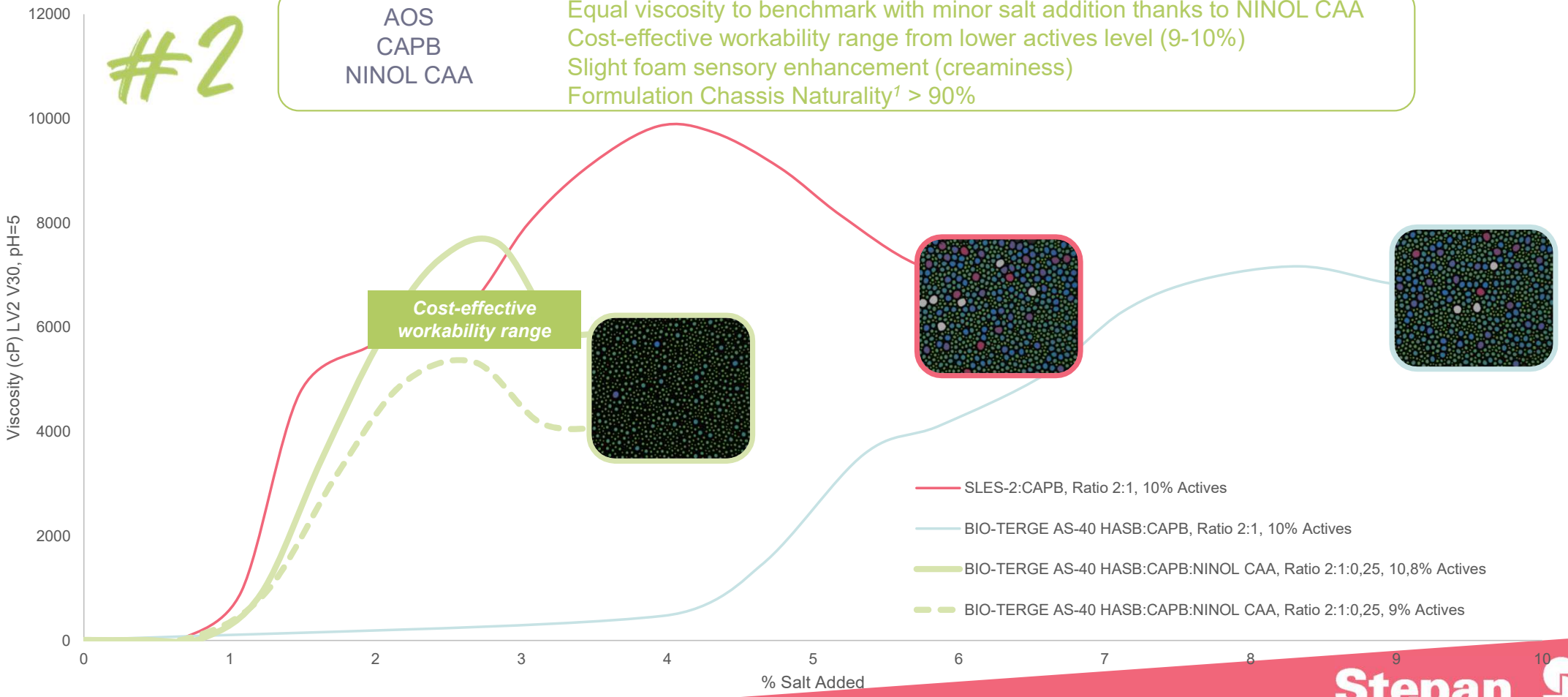
Sulfonate Thickening Hacks

Achieve similar performance to benchmark SLES-2/CAPB while **managing costs**

#2

AOS
CAPB
NINOL CAA

Equal viscosity to benchmark with minor salt addition thanks to NINOL CAA
Cost-effective workability range from lower actives level (9-10%)
Slight foam sensory enhancement (creaminess)
Formulation Chassis Naturality[1] > 90%



[1]*Formulation Chassis Naturality= % of the ingredients multiplied by Natural Origin Index/Content (NOI/NOC) Calculation according to ISO 16128*
Disclaimer: ISO 16128: Stepan interprets the Natural Origin Index (NOI) under ISO 16128 to be calculated based on the biorenewable carbon approach allowed by the norm.
We take water into account (NOI=1), additives and known impurities. Stepan assumes all raw materials and processes used to manufacture our products to be accepted by the norm.

Stepan
Personal Care

NINOL CAA, a naturally derived aesthetic enhancer



	NINOL CAA INCI: Dimethyl Lauramide/Myristamide
Claims	
Natural Origin Index ***ISO 16128 Score***[1]	0.86
Free From	DEA, Sulfates, EO, Preservatives, Polymers, Palm
Biodegradability	Readily
Mildness (Zein, 1-100 Scale, Lower is Better)	59
Actives %	100
Active Use Level %	0.25 – 2.0
Handling	Clear Liquid at RT
Cost Impact	Low to Medium
Lab Workup	Back-Add and Mix

[1] *Natural Origin Index/Content (NOI/NOC) Calculation according to ISO 16128*
Disclaimer: ISO 16128: Stepan interprets the Natural Origin Index (NOI) under ISO 16128 to be calculated based on the biorenewable carbon approach allowed by the norm.
We take water into account (NOI=1), additives and known impurities. Stepan assumes all raw materials and processes used to manufacture our products to be accepted by the norm.

Stepan
Personal Care

Non-ionic Secondary Surfactants

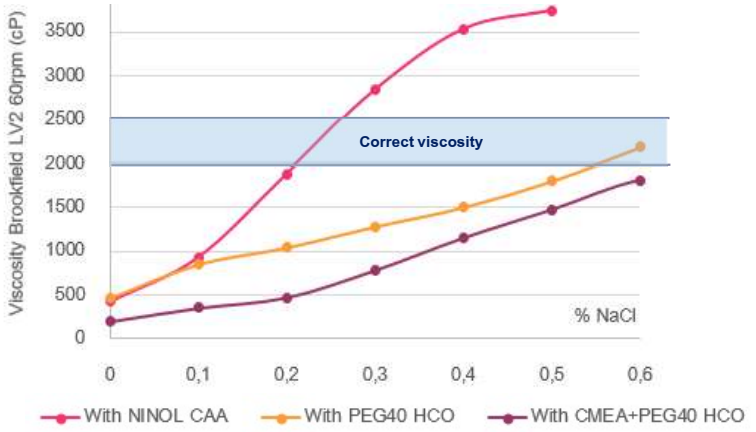
Optimize your formulation with high load of fragrance thanks to NINOL CAA.
Let's remove Cocamide MEA or Cocamide DEA from your formulation and enjoy high viscosities even with Sulfate-free

Ingredients	CMEA+PEG40HCO Chassis w/w%	PEG40 HCO Chassis w/w%	NINOL CAA Chassis. w/w %
SLES (25%)	40	40	36
CAPB	13,3	16,7	10,00
Cocamide MEA	1	/	/
Bambou fragrance	4	4	4
NINOL CAA	/	/	1
PEG-40 Hydrogenated Castor Oil	1	1	/
Water	40,7	38,3	49,0
Surfactant actives level	**15**	**15**	**13**

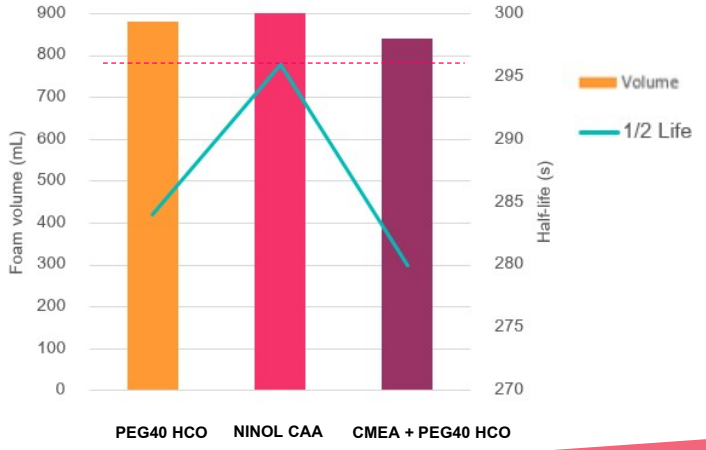
Removing CMEA

Replacing CMEA+ PEG40 HCO by NINOL CAA + actives reduction

Viscosity building



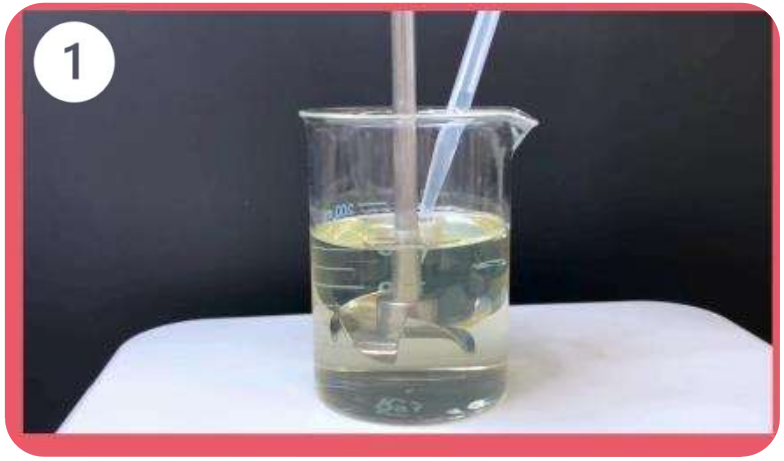
Foaming power

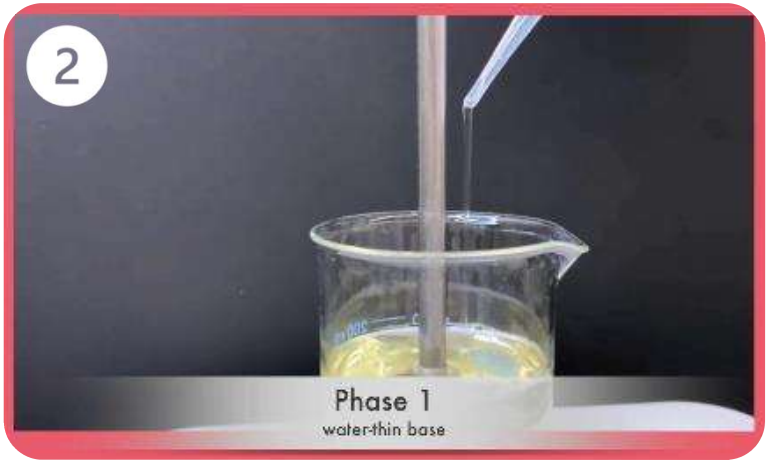


Stepan Personal Care

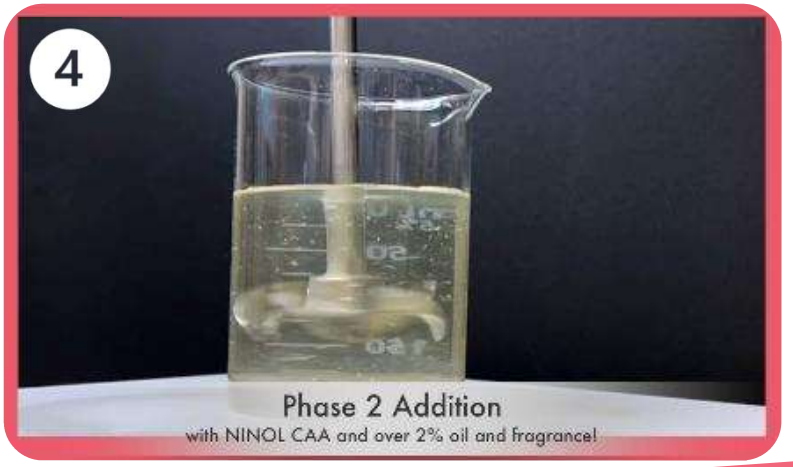
NINOL CAA Powerful Viscosity Builder

Opt for an Ingredient That Removes the Need for Salt











Stepan
Personal Care

Prototype Details
Boundless AOS Base

Stepan
Personal Care

Prototypes Design







BOUNDLESS BASE: PERSONAL CARE CLEANSING

Bounless base

Formulation N° 1610

Phase	Trade Name	INCI	% by weight
A	BIO-TERGE® AS-40 HP	Sodium C14-16 Olefin Sulfonate	53,3
	AMPHOSOL® HCG-HP	Cocamidopropyl Betaine	41,0
	DI Water	Aqua	q.s. to 100.00

Attributes	
Viscosity	45,000 – 75,000 cPs
Feel	Hazy to transparent viscous liquid



Stepan Personal Care

Economy Hand Soap

ELN Reference No. 5388-175-102

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	Covera™, Hallstar	Aloe Barbadensis Leaf Juice	1.00
	Glycerin	Glycerin	0.10
B	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	21.88
C	Microcare® BDB, Thor	Benzyl Alcohol (and) Benzoic Acid (and) Dehydroacetic Acid	0.50
	NINOL® CAA	Dimethyl Lauramide/Myristamide	0.25
D	Polysorbate 20	Polysorbate 20	0.18
	086610 Cucumber Citrus, Chemia	Fragrance	0.05
E	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.
	Sodium Chloride	Sodium Chloride	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP NINOL® CAA	7.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, Sodium Chloride, Dimethyl Lauramide/Myristamide, Glycerin, Aloe Barbadensis Leaf Juice, Benzyl Alcohol, Benzoic Acid, Dehydroacetic Acid, Polysorbate 20, Fragrance, Citric Acid, Sodium Hydroxide

Attributes	
Viscosity	4,000 (3,000 – 5,000) cPs
Feel	Hollow and tacky upon drying
Foam	Light and fluffy



Stepan Personal Care

Prestige Hand Soap

ELN Reference No. 5388-175-103

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	Glycerin	Glycerin	0.10
B	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	29.49
	STEPAN-MILD® LSB	Sodium Lauryl Sulfoacetate (and) Disodium Laureth Sulfosuccinate	8.00
C	SymOcide® PDO, Symrise	Phenoyxethanol (and) Caprylyl Glycol	0.75
	STEPAN-MILD® GCC	Glyceryl Caprylate/Caprate	0.25
	Glycerox™ HE, Croda	PEG-7 Glyceryl Cocoate	1.00
D	Beta Glucan, Making Cosmetics	Water (and) Glycerin (and) Beta-Glucan	0.01
	Ajidew® N-50, Ajinomoto	Sodium PCA (and) Aqua	0.01
	HerbaMaker™ Hydro, Making Cosmetics	Glycerin, Aloe Barbadensis Leaf Juice, Cucumis Sativus (Cucumber) Fruit Extract, Saccharum Officinarum (Sugar Cane) Extract, Helianthus Annuus (Sunflower) Petals Extract, Chondrus Crispus (Irish Moss) Extract, Sodium Hyaluronate, Water (Aqua)	0.01
	Vitamin E Acetate	Vitamin E Acetate	0.01
E	Polysorbate 20	Polysorbate 20	2.00
	086634 Sea Salt Ozone, Chemia	Fragrance	1.00
F	Synthalen® W400, 3V Sigma USA Inc.	Acrylates Copolymer	5.50
G	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.
	Sodium Chloride	Sodium Chloride	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP STEPAN-MILD® LSB STEPAN-MILD® GCC	12.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, Sodium Lauryl Sulfoacetate, Disodium Laureth Sulfosuccinate, PEG-7 Glyceryl Cocoate, Acrylates Copolymer, Glyceryl Caprylate/Caprate, Sodium Chloride, Vitamin E Acetate, Glycerin, Aloe Barbadensis Leaf Juice, Cucumis Sativus (cucumber) fruit extract, Saccharum Officinarum (sugarcane) extract, Helianthus Annuus (sunflower) petals extract, Chondrus Crispus (Irish moss) extract, Cucumis Sativus (cucumber) fruit extract, Saccharum Officinarum (sugarcane) extract, Helianthus Annuus (sunflower) petals extract, Chondrus Crispus (Irish moss) extract, Phenoxyethanol, Caprylyl Glycol, Polysorbate 20, Fragrance, Citric Acid, Sodium Hydroxide



Attributes

Viscosity	7,500 (5,500 – 9,500) cPs
Feel	Longer rinseability Smooth after-feel
Foam	Lots of flash foam and volume Large bubble structures Smooth lather

Stepan®
Personal Care

Economy Body Wash

ELN Reference No. 5388-175-104

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	Covera™, Hallstar	Aloe Barbadensis Leaf Juice	1.00
	Glycerin	Glycerin	0.10
B	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	31.25
C	Microcare® PE, Thor	Phenoyethanol	0.75
	NINOL® CAA	Dimethyl Lauramide/Myristamide	0.25
D	Polysorbate 20	Polysorbate 20	0.38
	086609 Lavender Aloe, Chemia	Fragrance	0.13
E	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.
	Sodium Chloride	Sodium Chloride	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP NINOL® CAA	10.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, Sodium Chloride, Dimethyl Lauramide/Myristamide, Glycerin, Aloe Barbadensis Leaf Juice, Phenoxyethanol, Polysorbate 20, Fragrance, Citric Acid, Sodium Hydroxide

Attributes	
Viscosity	7,000 (5,250 – 8,750) cPs
Feel	Hollow and moisturizing
Foam	Light and fluffy



Stepan Personal Care

Prestige Body Wash

ELN Reference No. 5388-175-105

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	Glycerin	Glycerin	2.00
B	Sodium Cocoyl Isethionate	Sodium Cocoyl Isethionate	3.82
	LATHANOL® LAL	Sodium Lauryl Sulfoacetate	3.85
C	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	28.91
D	Hydrolite® 8 Green, Symrise	Caprylyl Glycol	0.50
	STEPAN-MILD® GCC	Glyceryl Caprylate/Caprate	0.10
	SymOcide® PDO, Symrise	Phenoyxethanol (and) Caprylyl Glycol	1.00
E	Beta Glucan, Making Cosmetics	Water (and) Glycerin (and) Beta-Glucan	0.01
	Ajidew® N-50, Ajinomoto	Sodium PCA (and) Aqua	0.01
	HerbaMaker™ Hydro, Making Cosmetics	Glycerin, Aloe Barbadensis Leaf Juice, Cucumis Sativus (Cucumber) Fruit Extract, Saccharum Officinarum (Sugar Cane) Extract, Helianthus Annuus (Sunflower) Petals Extract, Chondrus Crispus (Irish Moss) Extract, Sodium Hyaluronate, Water (Aqua)	0.01
	Vitamin E Acetate	Vitamin E Acetate	0.01
F	Polysorbate 20	Polysorbate 20	1.00
	217435 Mountain Ozone, Belle Aire	Fragrance	1.00
G	PEG-120 Methyl Glucose	PEG-120 Methyl Glucose	1.25
H	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP STEPAN-MILD® GCC LATHANOL® LAL	15.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, PEG-120 Methyl Glucose, Sodium Lauryl Sulfoacetate, Sodium Cocoyl Isethionate, Glyceryl Caprylate,Caprate, Cucumis Sativus (Cucumber) fruit extract, Saccharum Officinarum (sugarcane) extract, Helianthus Annuus (sunflower) Petals Extract, Chondrus Crispus (Irish Moss) Extract, Avena Sativa (oat) Meal Extract, Sodium Hyaluronate, Vitamin E Acetate,Glycerin, Caprylyl Glycol, Aloe Barbadensis Leaf Juice, Phenoxyethanol, Polysorbate 20, PEG-120 Methyl Glucose, Fragrance, Citric Acid, Sodium Hydroxide

Attributes	
Viscosity	10,000 (7,000 – 13,000) cPs
Feel	Clean and smooth
Foam	Lots of flash foam and volume Large bubble structures



Stepan® Personal Care

Economy Shampoo

ELN Reference No. 5388-175-106

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	UCARE™ Polymer JR-125, Dow	Polyquaternium-10	0.20
B	Glycerin	Glycerin	0.10
	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	31.25
	Microcare® BDB, Thor	Benzyl Alcohol (and) Benzoic Acid (and) Dehydroacetic Acid	0.75
	NINOL® CAA	Dimethyl Lauramide/Myristamide	0.25
C	Polysorbate 20	Polysorbate 20	0.36
	ORC2400684 Strawberry Peony, Orchidia	Fragrance	0.12
D	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.
	Sodium Chloride	Sodium Chloride	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP NINOL® CAA	10.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, Sodium Chloride, Dimethyl Lauramide/Myristamide, Glycerin, Polyquaterinum-10, Benzyl Alcohol, Benzoic Acid, Dehydroacetic Acid, Polysorbate 20, Fragrance, Citric Acid, Sodium Hydroxide

Attributes	
Viscosity	5,000 (3,000 – 7,000) cPs
Feel	Hollow and moisturizing
Foam	Light and fluffy



Stepan Personal Care

Prestige Shampoo

ELN Reference No. 5388-175-107

Phase	Trade Name	INCI	% by weight
A	Water	Water	q.s. to 100.00
	UCARE™ Polymer JR-125, Dow	Polyquaternium-10	0.20
B	Glycerin	Glycerin	0.10
	LATHANOL® LAL	Sodium Lauryl Sulfoacetate	7.00
	Decyl Glucoside	Decyl Glucoside	2.94
	HALLSTAR® EGMS, Hallstar	Glycol Stearate	1.00
	HALLSTAR® EGDE, Hallstar	Glycol Distearate	0.50
	STEPANQUAT® Helia	Disunfloweroylethyl Dimonium Chloride (and) Sunflower Seed Oil Glycerides (and) Lauryl Lactyl Lactate	1.00
C	Boundless Base – No. 1610 BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP	Sodium C14-16 Olefin Sulfonate (and) Cocamidopropyl Betaine	38.33
	Microcare® BDB, Thor	Benzyl Alcohol (and) Benzoic Acid (and) Dehydroacetic Acid	0.75
D	Polysorbate 20	Polysorbate 20	2.00
	SZUSM0000643 Rosemary Lavender, Chemia	Fragrance	1.00
E	PEG-120 Methyl Glucose	PEG-120 Methyl Glucose	5.00
F	Citric Acid	Citric Acid	q.s.
	Sodium Hydroxide	Sodium Hydroxide	q.s.
	Sodium Chloride	Sodium Chloride	q.s.

Essential Surfactants	Total Actives %
BIO-TERGE® AS-40 HP AMPHOSOL® HCG-HP LATHANOL® LAL STEPANQUAT® Helia	15.0

Ingredient Listing

Water, Sodium C14-16 Olefin Sulfonate, Cocamidopropyl Betaine, Sodium Lauryl Sulfoacetate, Decyl Glucoside, PEG-120 Methyl Glucose, Disunfloweroylethyl Dimonium Chloride, Sunflower Seed Oil Glycerides, Lauryl Lactyl Lactate, Glycol Stearate ,Glycol Distearate, Glycerin, Polyquaterinum-10, Benzyl Alcohol, Benzoic Acid, Dehydroacetic Acid, Polysorbate 20, Fragrance, Citric Acid, Sodium Hydroxide

Attributes	
Viscosity	2,500 (2,000 – 5,000) cPs
Feel	Clean and smooth Moisturizing
Foam	Dense and creamy



Stepan Personal Care

Comprehensive Evaluation of Stepan Additives

Stepan
Personal Care

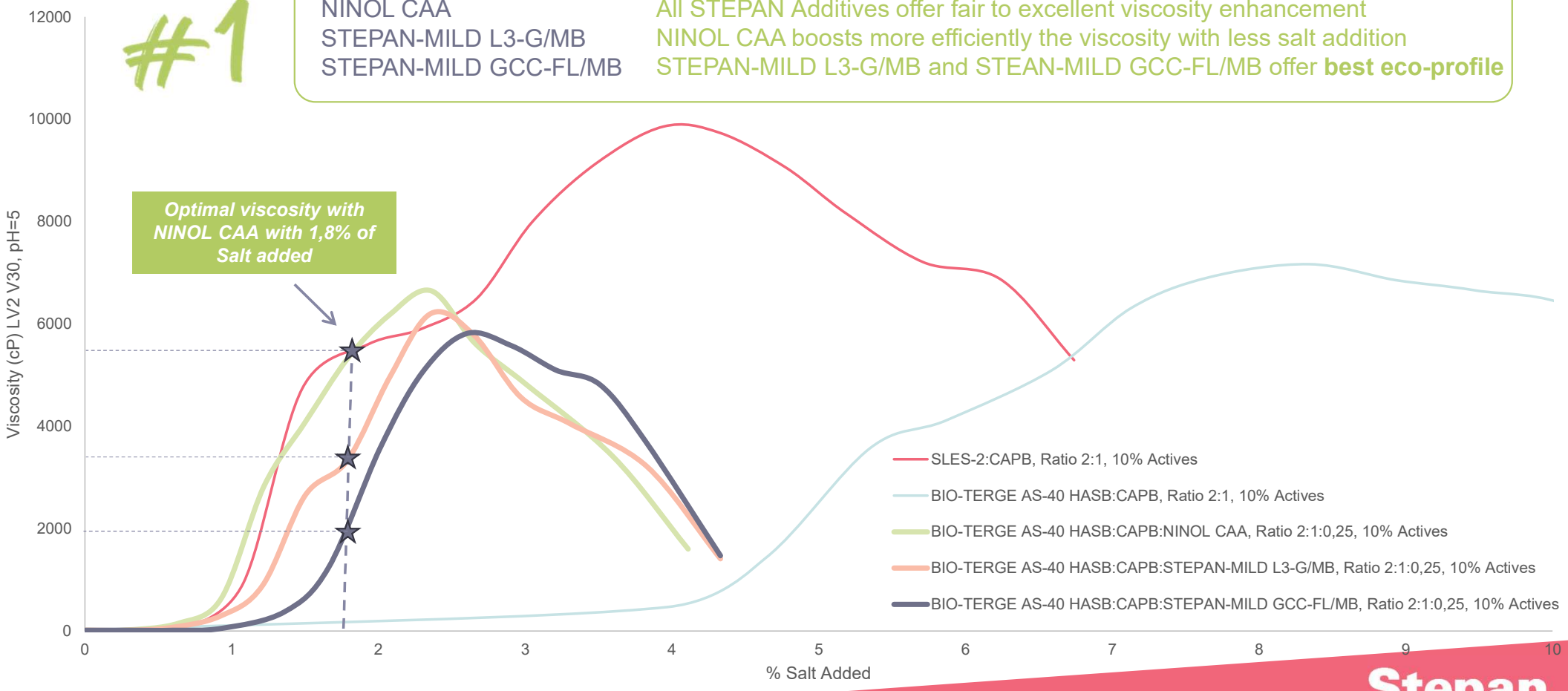
Sulfonate Thickening Hacks with STEPAN additives

Hack 1: NINOL CAA compared to Cosmos Approved Emollients

#1

NINOL CAA
STEPAN-MILD L3-G/MB
STEPAN-MILD GCC-FL/MB

All STEPAN Additives offer fair to excellent viscosity enhancement
NINOL CAA boosts more efficiently the viscosity with less salt addition
STEPAN-MILD L3-G/MB and STEAN-MILD GCC-FL/MB offer **best eco-profile**



Stepan
Personal Care

Specialty Additive Surfactants

Optimize choices for improving viscosity while enhancing eco-profile

Whether you are looking for sensory enhancement, conditioning effects or an enriched experience, these surfactant additives are Cosmos Approved and can add multifunctionality to your formulation.

Verified by ECOCERT
COSMOS APPROVED

STEPAN-MILD® GCC Series

STEPAN-MILD GCC/MB[1]

INCI Name	Glyceryl Caprylate/Caprate
Appearance	Paste
Active	100%
Preservative	None

STEPAN-MILD GCC-FL/MB

INCI Name	Glyceryl Caprylate/Caprate (and) Glycerin (and) Propanediol
Appearance	Liquid Viscous
Active	62%
Preservative	None



Foam booster
Skin feel enhancer
Conditioning properties

BCI=100
Readily biodegradable
No preservative
Cosmos certified

Easily thickens sulfate-free systems
Easy handling and formulating
Achieve water-white formulas

STEPAN-MILD L3-G/MB

INCI Name	Lauryl Lactyl Lactate
Appearance	Clear Liquid
Active	100%
Preservative	None



Enhances mildness
Enhance emolliency
Conditioning properties

BCI=100
Readily biodegradable
No preservative
Cosmos Certified

Easily thickens sulfate-free systems
Easy handling and formulating
Achieve water-white formulas

[1]MB designation represents RSPO Mass Balance-certified ingredients

Stepan Personal Care

STEPAN Additives: A Matter of Choice

Options to achieve your ideal equilibrium

Legend

★★★ Excellent
★★ Good
★ Marginal
N/A Not applicable

Stepan Offering	STEPAN-MILD GCC/MB			STEPAN-MILD GCC-FL/MB			STEPAN-MILD L3-G/MB		NINOL CAA
INCI Name	Glyceryl Caprylate/Caprate			Glyceryl Caprylate/Caprate (and) Glycerin (and) Propanediol			Lauryl Lactyl Lactate		Dimethyl Lauramide/Myristamide
Product Form	100% active, soft paste			62% active, viscous liquid			100% active, liquid		100% active, liquid
NOI/NOC[1]	1			1			1		0,86
Viscosity Building	★★			★★			★★		★★★
Foam Volume	★★★			★★★			★★		★★★
Foam Creaminess	★★			★★			★★★		★★★
Foam Stability	★★			★★			★★★		★★★
Conditioning Booster	★★★ Synergy with polyquaternium			★★★ Synergy with polyquaternium			★★★ Silicone alternative for 2-in-1 shampoo		N/A
Preservative Booster	★★★			★★★			N/A		N/A
Active/Fragrance Solubilization	N/A			N/A			★★★ Eg. Salicylic acid		★★★ Fragrances
Structured Liquid System	★★★			★★★			N/A		N/A
Cold Process	N/A			★★★			★★★		★★★
Emolliency	★★★			★★★			★★★		N/A
Lubricity	★			★			★★★		★★★
Sensory Profile[2]	Melting butter texture and rich, cushioned feel			Fluid, cushioned feel			Light, fresh, fluid feel and satin finish		N/A
Claims	Cosmos	Sulfate-free	China Compliant	Cosmos	Sulfate-free	China Compliant	Cosmos	Sulfate-free	Sulfate-Free

[1]*Natural Origin Index/Content (NOI/NOC) Calculation according to ISO 16128*
Disclaimer: ISO 16128: Stepan interprets the Natural Origin Index (NOI) under ISO 16128 to be calculated based on the biorenewable carbon approach allowed by the norm.
We take water into account (NOI=1), additives and known impurities. Stepan assumes all raw materials and processes used to manufacture our products to be accepted by the norm.
[2]*Sensory analysis on pure ingredient completed by 20 expert panelists. Contact Stepan for more information.*

Stepan® Personal Care



Stepan
Personal Care

Stepan
Personal Care

Formulation Inspirations



Alpha Olefin Sulfonate, You Rock!

There are countless formulation options available… Eenie, Meenie, Miney Moe.

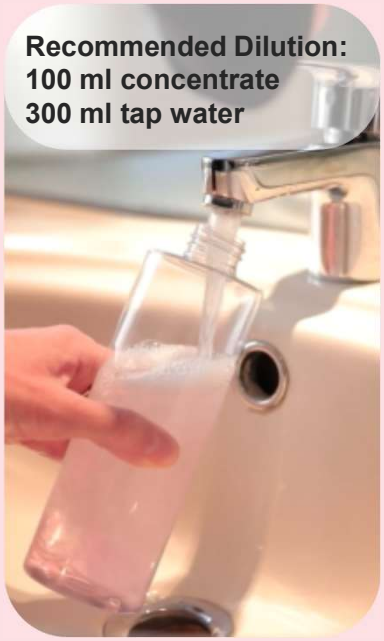
Oh Honey! Honey Time Shampoo

Formulation No. 1492



Concentrated to dilute

Formulation No. 1443



Choco Delight: Powder Shampoo

Formulation No. 1488



Stepan Offers Global Development Support



Northfield, IL
(Headquarters)

Voreppe,
France





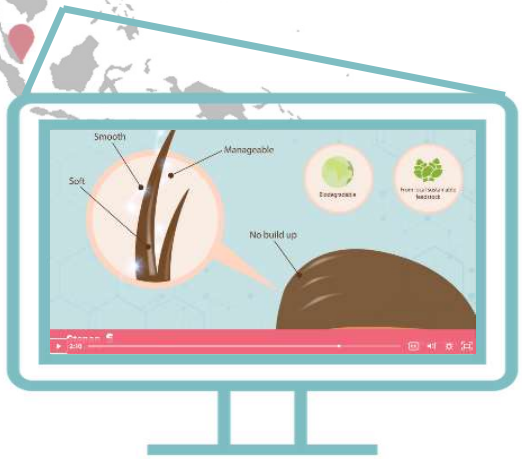
Mexico City /
Ecatepec, Mexico

Bogotá,
Colombia

São
Paulo,
Brazil



Jurong
Island,
Singapore





Stepan
Personal Care



Stepan
Personal Care

Thank you!

Visit stepan.com/personalcare
for more information.

Follow us on:

2025 © Stepan Company. All Rights Reserved.
ALPHA-STEP®, AMPHOSOL®, BIO-TERGE®, MAPROSYL®, NINOL®, STEPANBLEND® and STEPAN-MILD® are registered trademarks of Stepan Company.
The information contained herein is based on the manufacturer's own study and the works of others and is subject to change without prior notice. The information is not intended to be all-inclusive, including as to the manner and conditions of use, handling, storage or disposal or other factors that may involve additional legal, environmental, safety or performance considerations. Nothing contained herein grants or extends a license, express or implied, in connection with any patents issued or pending of the manufacturer or others, or shall be construed as a recommendation to infringe any patents. STEPAN COMPANY MAKES NO PRODUCT WARRANTY OF MERCHANTABILITY OR OF FITNESS FOR A PARTICULAR USE, EXPRESS OR IMPLIED, AND NO OTHER WARRANTY OR GUARANTY, EXPRESS OR IMPLIED, IS MADE, INCLUDING AS TO INFORMATION REGARDING PERFORMANCE, SAFETY, SUITABILITY, STABILITY, ACCURACY, COMPLETENESS, OR ADEQUACY. Stepan Company (and its employees, subsidiaries and affiliates) shall not be liable (regardless of fault) to the vendee, its employees, or any other party for any direct, indirect, special or consequential damages arising out of or in connection with the accuracy, completeness, adequacy, furnishing, use, or reliance upon information provided herein. The vendee assumes and releases Stepan Company (and its employees, subsidiaries and affiliates) from all liability, whether in tort, contract or otherwise to the fullest extent possible under the relevant law.